

MB Note

1 Population Models for Single Species

Logistic Growth Model (Population): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K})$ N: population at time t ; K: carrying capacity (depends on resources); $r, K > 0$.

Sol: $N(t) = \frac{N_0 K e^{rt}}{K + N_0(e^{rt} - 1)} \xrightarrow{\infty} K$ **Equilibrium points:** $N = 0$ (unstable), $N = K$ (stable). **Quickest:** $N = \frac{K}{2}$ (<: 凹; >: 凸)

General Population Model: Model: $\frac{dN}{dt} = f(N)$ $f(N)$: nonlinear function. **No limit cycle periodic sol.**

· N^* equilibrium point ($f(N^*) = 0$) $\Rightarrow$ $f'(N^*) < 0$ (stable for small perturbations); $f'(N^*) > 0$ (unstable) **Time-Independent:** Equilibrium | $\frac{dN}{dt} = 0 \Rightarrow N(t) = N(t - T) = N^*$.

Linearise at N^* : Consider $N(t) = N^* + n(t)$, $|n(t)| \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} = f'(N^*)n(t) = f'(N^*)n(t)$ (Delay) 带入 $f \approx$ (求 2 次以上) **Small Perturbations:** $n(t)$

Stability N^* : 计算扰动 $n(t)$ 的 $\frac{dn}{dt} \rightarrow$ 假设 $n(t) = e^{\lambda t}$ | $ce^{\lambda t}$ 求解 λ 隐式表达 $\rightarrow \lambda = u + iw$ 带入解出 u 和 w . $\rightarrow u < 0$ asy:stabl; $u > 0$ unstable. **Find T_c :** 令 (delay) $T = 0$, 验稳定性, 如稳, $u = 0$ ($\lambda = iw$) 为 T_c

Delay Model: Model: $\frac{dN}{dt} = f(N(t), N(t - T))$ T: delay time > 0 周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

¹Ext. of Logistic Model: $\frac{dN}{dt} = rN(t) \left[1 - \frac{N(t-T)}{K} \right]$ (用 "ave. pop." 估计, 准用 "Convolution Type") $\Rightarrow$ ²Dimensionless: $\frac{dN^*}{dt^*} = N^*(t^*) \left[1 - N^*(t^* - T^*) \right]$ ($N^* = \frac{N}{K}$, $t^* = rt$, $T^* = rT$)

^{1,2}Periodic sol: stable limit cycle $N(t + t_p) = N(t)$, t_p : period. **Amplitude:** rT **Equilibrium Points:** $N = 0$ | $N^* = 0$ (unstable); $N = K$ | $N^* = 1$ (Stable: $0 < T^* = rT < \frac{\pi}{2}$; Unstable: $T^* > \frac{\pi}{2}$)

Continuous Population Model (EN): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - EN = f(N)$ $r, K, E > 0$; EN: harvesting rate/unit time; E: effort measure.

Equilibrium: $N = 0$ (unstable); $N = N_h(E) = K(1 - \frac{E}{r}) > 0$ (need $E < r$, Stable) **Yield:** $Y(E) = EN$ · $E > r$: die out; else: **Max Yield:** $Y_M = \frac{rK}{4}$ ($E = \frac{r}{2}$) **Steady State:** $N_h|_{Y_M} = \frac{K}{2}$

Continuous Population Model (Y_0): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$ Y_0 : constant yield rate/unit time.

Max Yield: $Y_M = \frac{rK}{4}$ **Equilibrium:** $1 < Y_0 < Y_M$: Two positive points (unstable[小] and stable[大]). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

2 Discrete Population Models

Discrete Model: Model: $N_{t+1} = N_t F(N_t) = f(N_t)$ · e.g. If $F(N_t) = r > 0$: sol will decays|grows geometrically for $r < 1$ | $r > 1$.

Max|Min: It's expected has max at N_m ($N_{\max} = f(N_m)$; $\frac{df}{dN_t}|_{N_m} = 0$); then $N_{\min} = f(N_{\max}) = f^2(N_m)$; If $N_{\min} \leq 1$, **species extinction**. **Equilibrium:** $N^* = f(N^*)$

Eigenvalue: $\lambda = f'(N^*)$ $N^* > N_m$: $|\lambda| < 1$: stable (critical bifurcation); $|\lambda| = 1$: periodic (attracting (+: tangent, -: period-doubling) bifurcation); $|\lambda| > 1$: unstable. (chaotic)

Perturbations at N^* : A.(stable) 扰动 $\rightarrow 0$ by a1. $0 < \lambda < 1$ ↓ monotone; a2. $-1 < \lambda < 0$ decreasing oscillations; B.(unstable) 扰动 $\rightarrow \infty$ by b1. $\lambda > 1$ ↑ monotone; b2. $\lambda < -1$ increasing oscillations.

Periodic: ¹ Equilibrium: $f^p(N^*) = N^*$ (指迭代 p 次) is p -periodic for original eq. ² $\forall$ even p -periodic N^* with $\lambda = -1$ branches into $2p$ -periodic N^* ; 后续分支间隔越来越短.

Sarkovskii's Theorem: If a 3-periodic N^* exists, then $\forall p \geq 1$, p -periodic N^* exists. (i.e. chaos) **Verhulst:** $N_{t+1} = rN_t(1 - \frac{N_t}{K})$; $r, K > 0$. Better: **Ricker curve:** $N_{t+1} = N_t e^{r(1 - \frac{N_t}{K})}$

Linearise at N^* : Consider $N_t = N^* + n_t$, $|n_t| \ll 1 \rightarrow N^* + n_{t+1} = \dots = f(N^* + n_t) \approx f(N^*) + f'(N^*)n_t \rightarrow n_{t+1} = f'(N^*)n_t = \lambda n_t \rightarrow n_t = \lambda^t n_0$. 分析 $t \rightarrow \infty$

3 Appendix

$y' + p(x)y = q(x)$: **Integrating Factor:** $I(x) = e^{\int p(x)dx} \Rightarrow I(x)y = \int I(x)q(x)dx$

Binomial Series: $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots$ **Linerisation:** $f(x + \Delta x) = f(x) + f'(x)\Delta x + \dots$